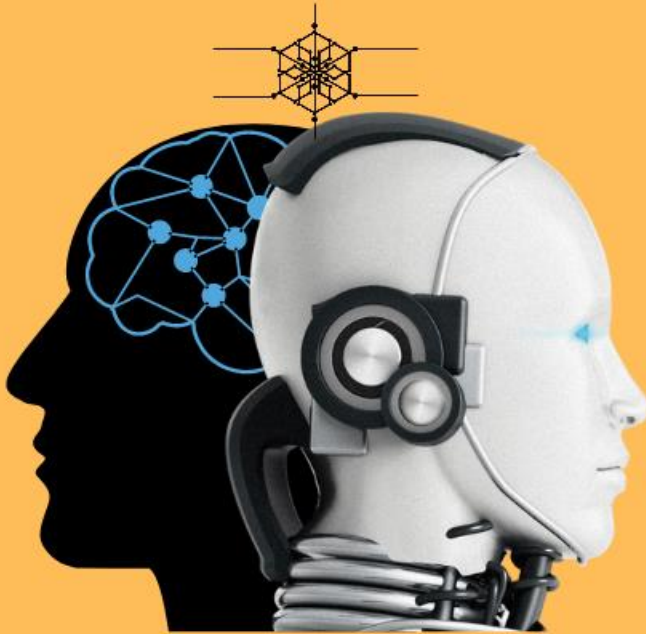


University of Stuttgart
Institute of Industrial Automation
and Software Engineering



Model-Based Systems Engineering and Adversarial Multi-Agent Reinforcement Learning

Name: Ahmed Mahfouz

Study Program: INFOTECH

Master Thesis final presentation



Motivation

Motivation

- **Motivation**
- Problem Statement
- Objectives
- Concept Design
- Prototype Development
- Experiments and Results
- Path Elimination Algorithm
- Conclusion and Future Work
- References

- Model-Based Systems Engineering (**MBSE**).
- Adversarial Multi-Agent Reinforcement Learning (**Adv-MARL**).



Figure 1: Model Based Systems Engineering (MBSE) [1].

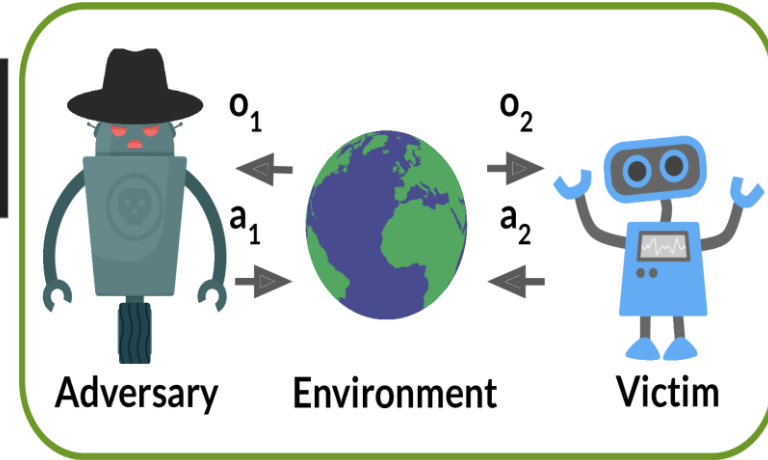


Figure 2: Multi-Agent Threat Model [2].

Problem statement

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- How can Adv-MARL principles be utilized to **convert DTMC** into **zero-sum games** to improve **safety** characteristics and decision-making in MBSE?

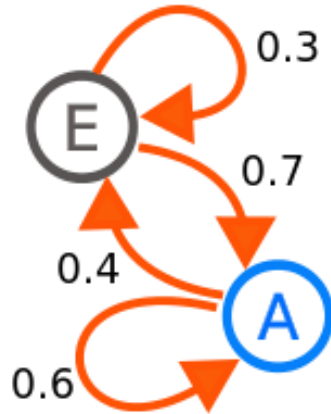


Figure 3: A Markov chain [3].



Figure 4: Zero sum game.

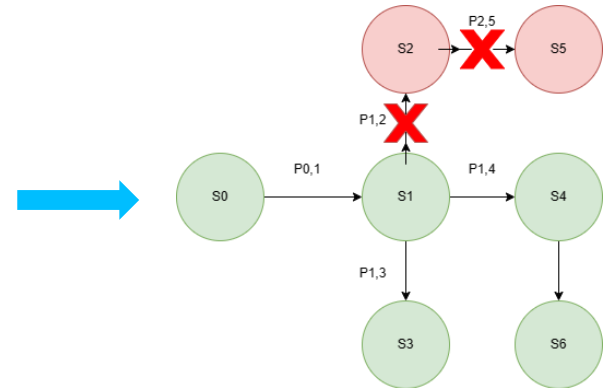


Figure 5: Path Elimination Algorithm .

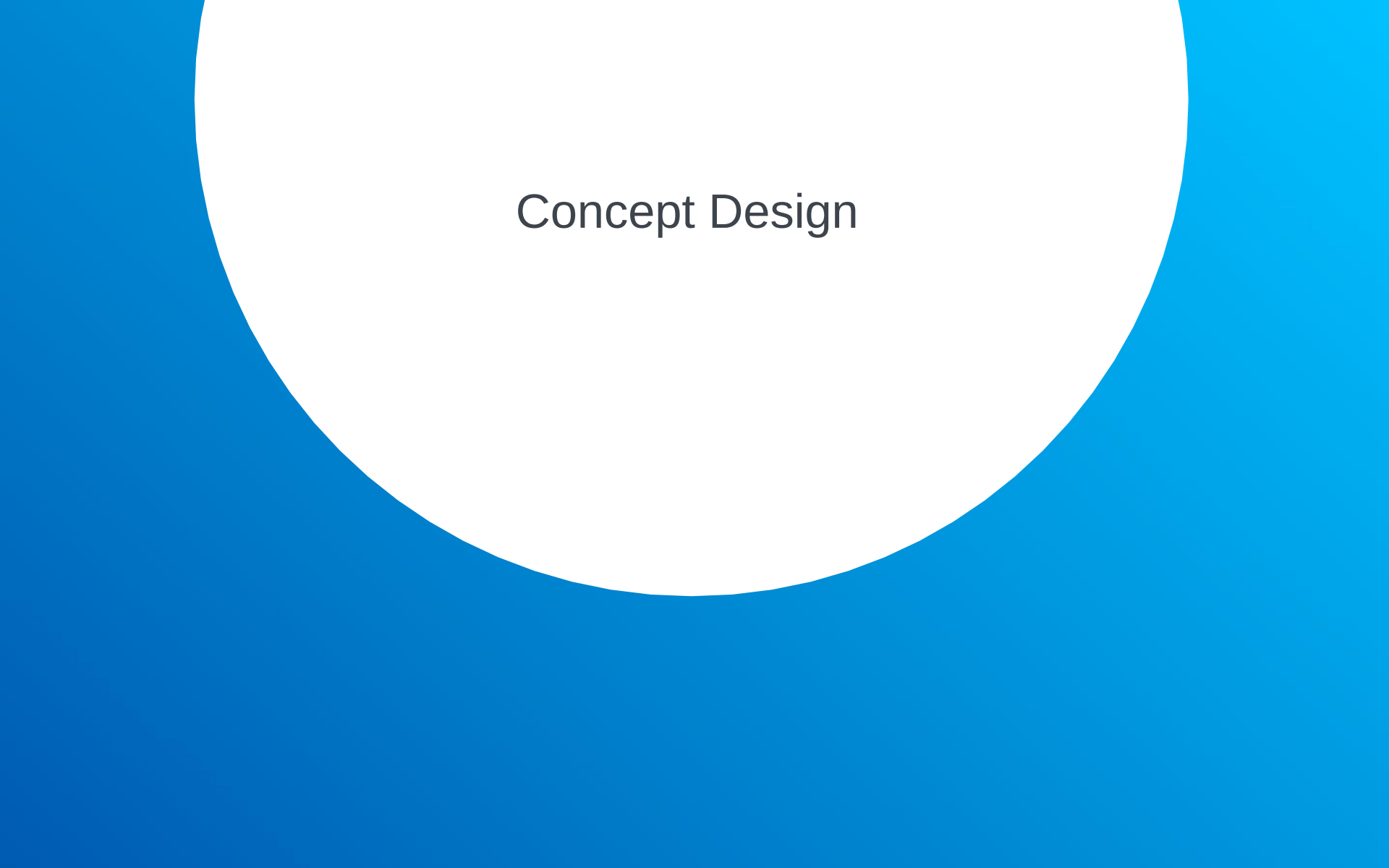
Outline

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Objectives

- ✓ Motivation
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- This project aims to:
 - Designing a **concept** for converting a **DTMC** into a **Zero-Sum** game.
 - Implementing a **prototype** for the developed concept.
 - Conducting different **experiments** to test the developed prototype under different conditions.
 - Develop a **path elimination** algorithm and introduce other algorithms and techniques from the literature.



Concept Design

Concept Design ● ○ ○

Game Description

✓ Motivation

✓ Problem
Statement

✓ Objectives

• **Concept Design**

• Prototype
Development

• Experiments
and Results

• Path
Elimination
Algorithm

• Conclusion and
Future Work

• References

- The game has **states** of the DTMC.
- Each state has **state undesirability**.
- The game has **two** agents (Blue Vs. Red).
- Each agent has initial **resources**.
- Each agent **action** has a **cost**.
- Agents take **turns** to perform actions.
- The game runs until one of the **termination conditions** is met.

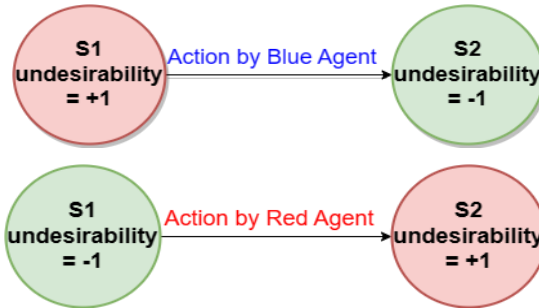


Concept Design ○ ● ○

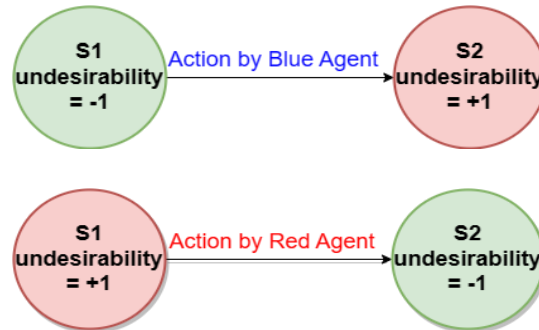
Agent Strategy

- ✓ Motivation
- ✓ Problem Statement
- ✓ Objectives
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- How does an agent **gain** points ?



- How does an agent **lose** points ?



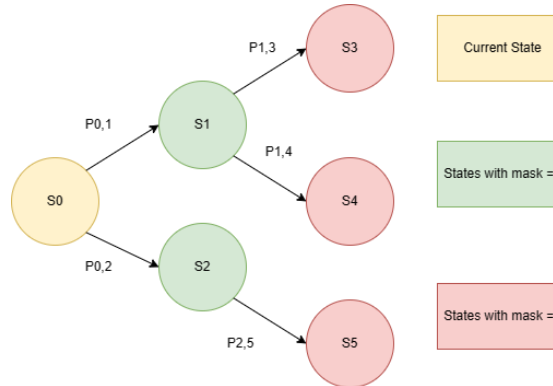
Concept Design ○ ○ ●

Environment Description

- ✓ Motivation
- ✓ Problem Statement
- ✓ Objectives
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- Observation Space:
 - A **discrete** observation space including all possible system states.
- Action Space:
 - A **discrete** space with all possible actions, each action represent a system state.

- Action Masking:



- Reward Function:

$$R = \Delta U = A(a) * [U(S_t) - U(S_{t+1})]$$

Where R is reward, A(a) is agent strategy, U(S) is state undesirability

Prototype Development

Prototype ● ○ ○

System Software Architecture

- ✓ Motivation
- ✓ Problem Statement
- ✓ Objectives
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- **Prototype Development**
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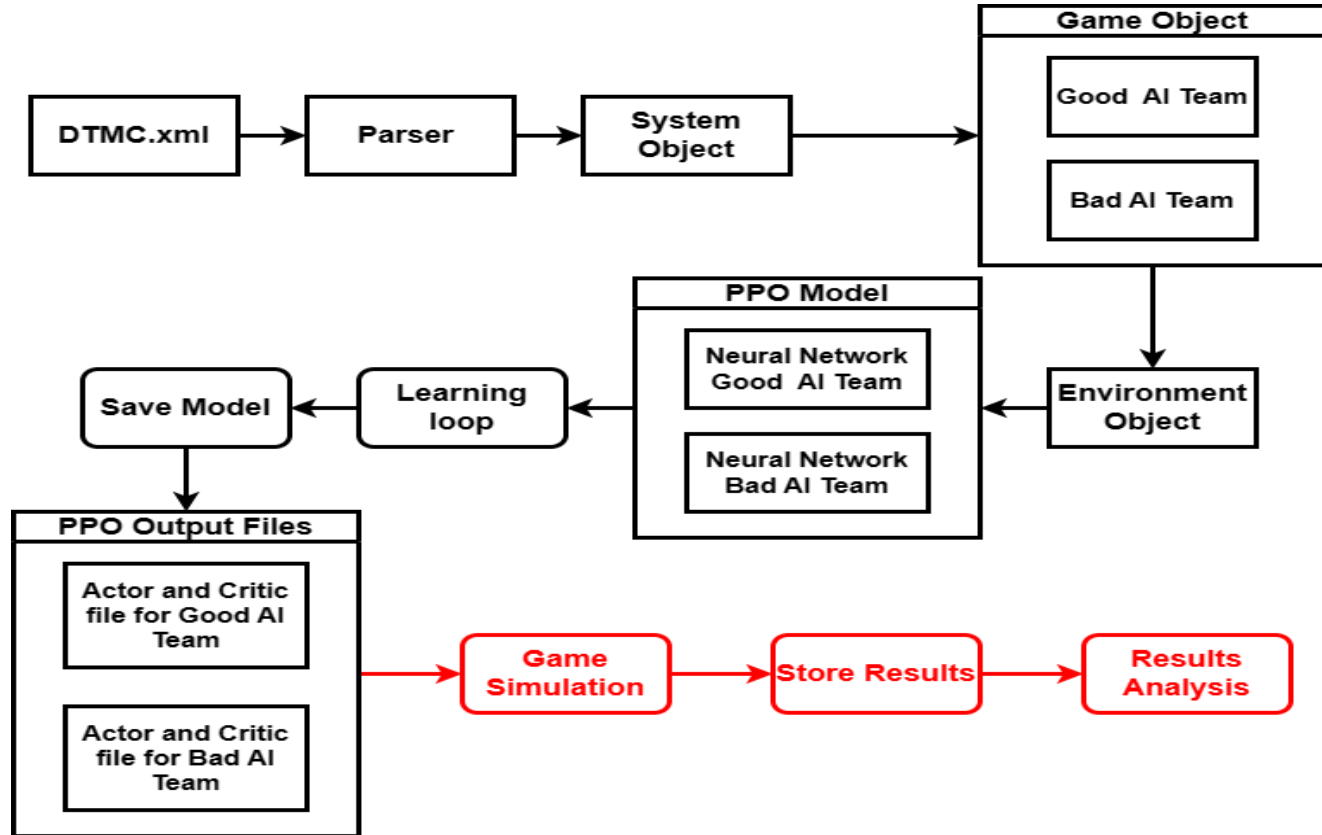


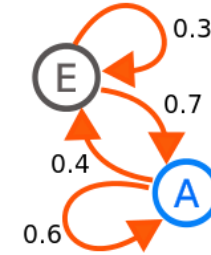
Figure 5: System Software Architecture [4].

Prototype ○ ● ○

Software Components ● ○

- ✓ Motivation
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- Parser Object:
 - **Parse** DTMC models from XML file into a system object.
- System Object:
 - Abstracts DTMC **attributes** and **dynamics**, such as system name, type, **states**, and **edges**.
- Game Object:
 - Created using the system object, it manages game **attributes**, players, and **dynamics**.

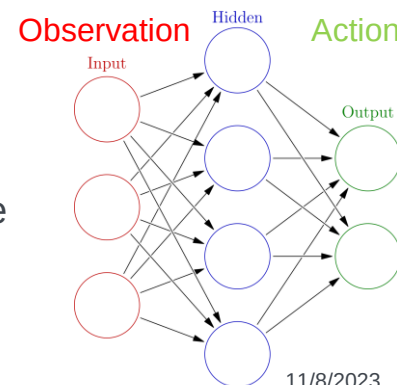
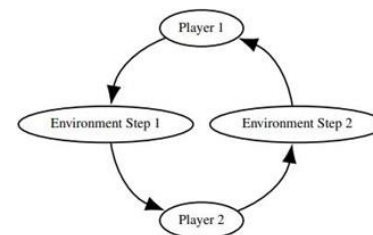
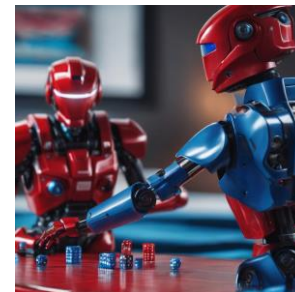


Prototype ○ ○ ●

Software Components ○ ●

- ✓ Motivation
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- Player Object:
 - Abstracts player **attributes**, such as name, score, and **strategies**.
- Environment Object:
 - Created based on the **PettingZoo** library **AECEnv**, sets up the environment for training.
- PPO-Model:
 - A **customized** PPO model to train the environment agents.
 - It was implemented with two **Neural Networks**, one for each agent.



Experiments and Results

Experiment and Results ● ○ ○

Experiment: Undesirability threshold

- ✓ Motivation
- ✓ Problem Statement
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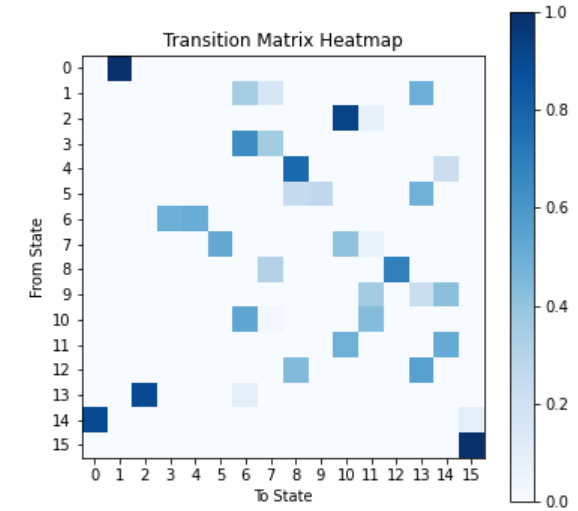
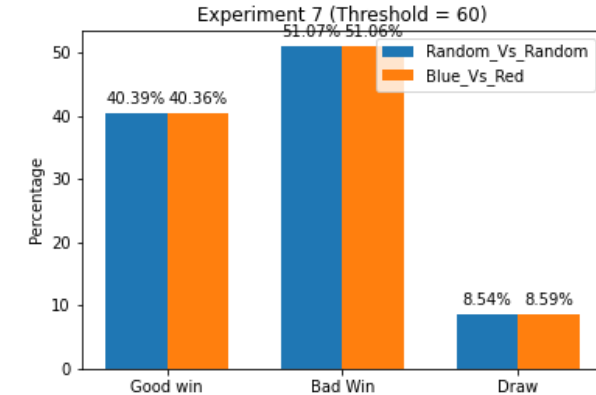
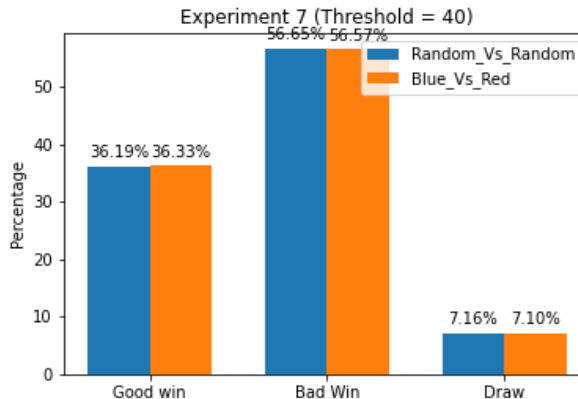
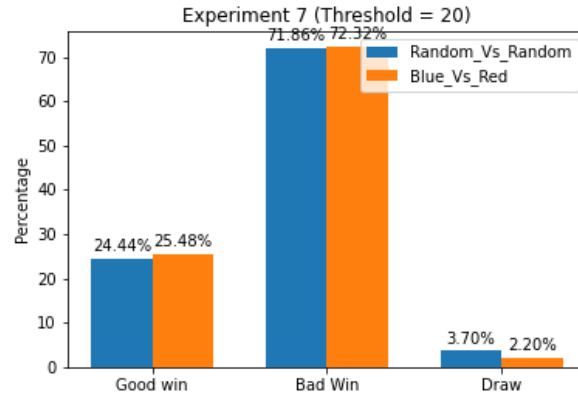


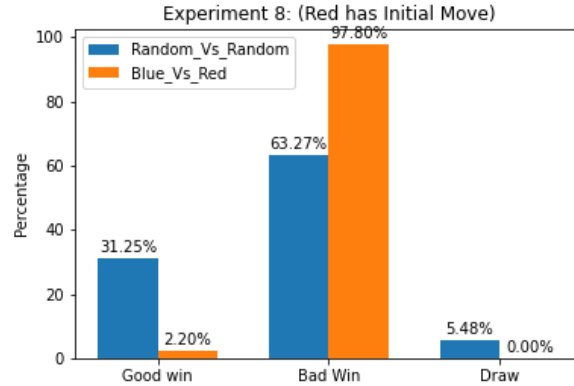
Figure 6: Results of Experiment: Undesirability threshold [4].

Experiment and Results ○ ● ○

Experiment: Changing Move Order

- ✓ Motivation
- ✓ Problem Statement
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Red has initial move.



Blue has initial move.

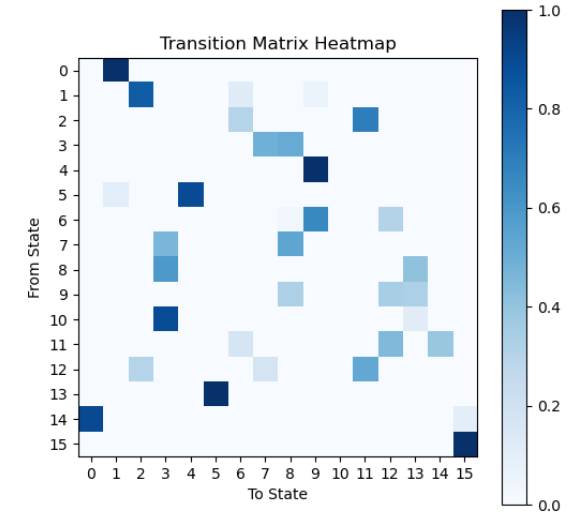
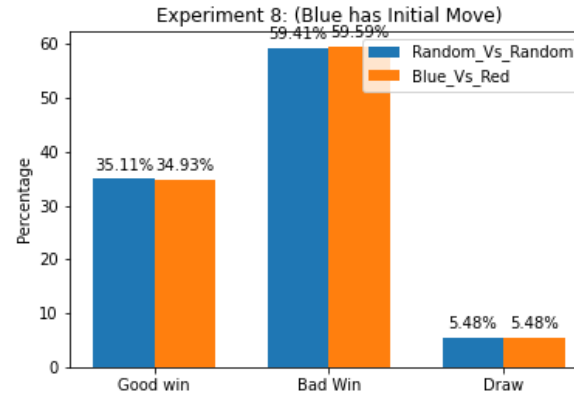


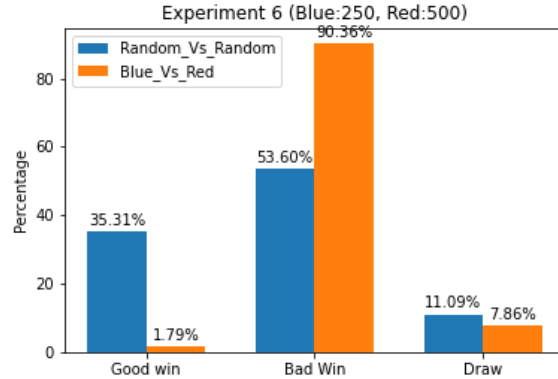
Figure 7: Results of Experiment: Changing Move Order[4].

Experiment and Results ○ ○ ●

Experiment: Initial Resources

- ✓ Motivation
- ✓ Problem Statement
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Initial resources: Blue = 250, Red = 500



Initial resources: Blue = 500, Red = 250

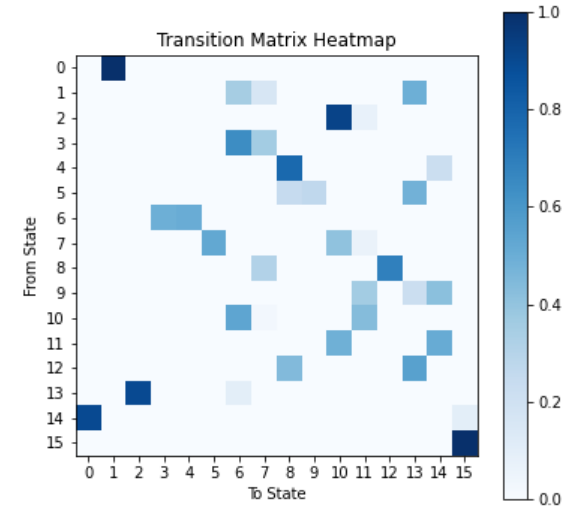
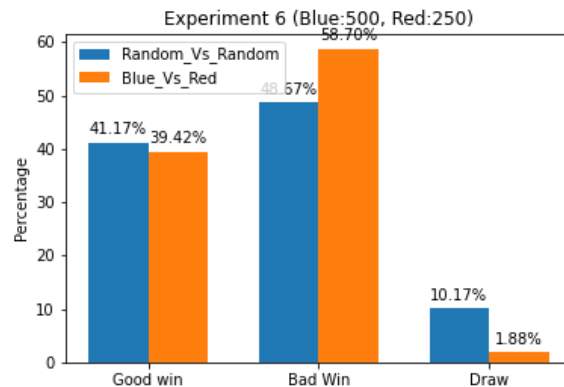


Figure 8: Results of Experiment: Initial Resources [4].

Path Elimination Algorithm

Path Elimination Algorithm ● ○

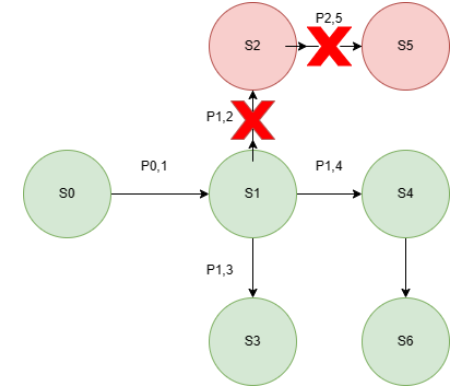
Methodology

- ✓ Motivation
- ✓ Problem Statement
- ✓ Objectives
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- Optimization Algorithm:
 - **Calculate** the initial **resource** allocated for each agent, so improbable paths are eliminated.
- Optimization Problem:

$$\begin{aligned} \max_x \quad & f(x) = \sum_{i=1}^{n=2} x_i \\ \text{subject to:} \quad & g_1(x) = \sum_{i=1}^{n=2} x_i \leq l_1 \\ & g_2(x) = x_2 \leq l_2 \end{aligned}$$

Where x_1 , and x_2 are initial resources for the Blue and Red agents, respectively.

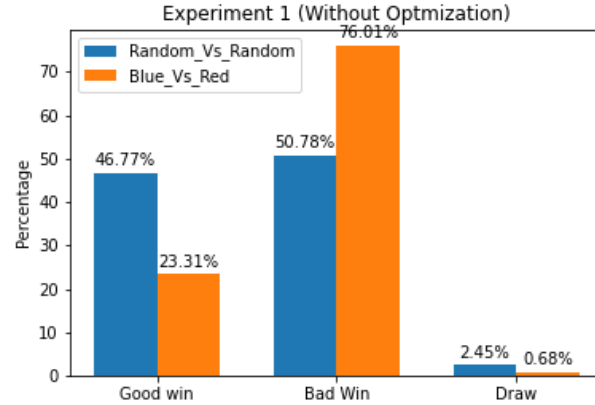


Path Elimination Algorithm ○ ●

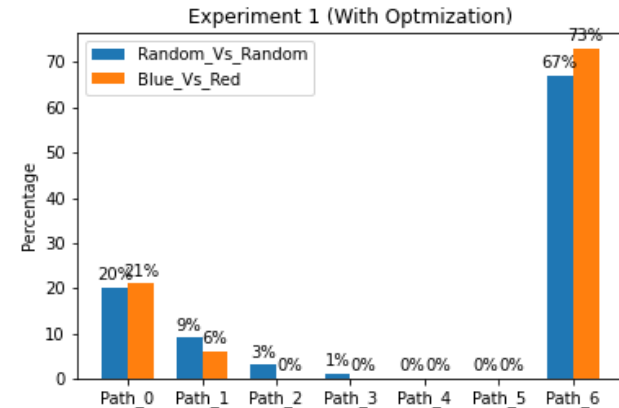
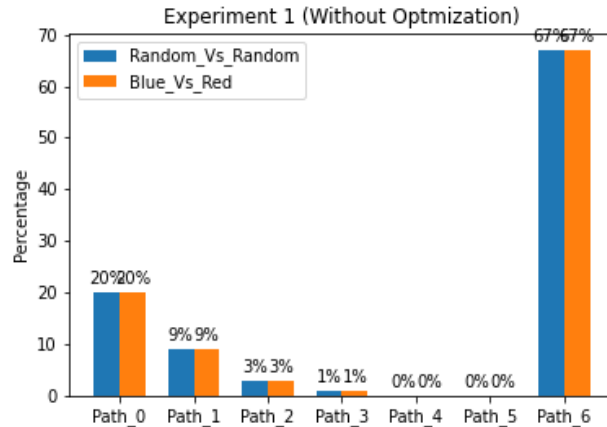
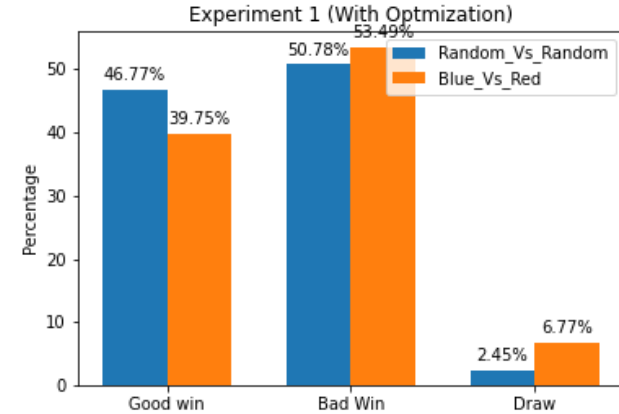
Results

- ✓ Motivation
- ✓ Problem Statement
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Initial resources: Blue = 1000, Red = 1000



Initial resources: Blue = 200, Red = 110



Conclusion and Future Work ● ○

- ✓ Motivation
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- Conclusion:

- A **concept** for converting a DTMC into Zero-Sum game was developed.
- A **prototype** implementation was developed for the concept.
- Different **experiments** were conducted to verify the concept and test the prototype.
- A **path elimination** algorithm to restrict action selection of the agents was developed.

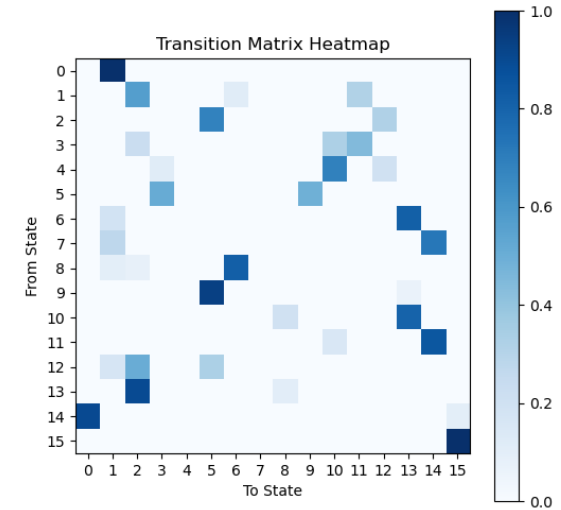
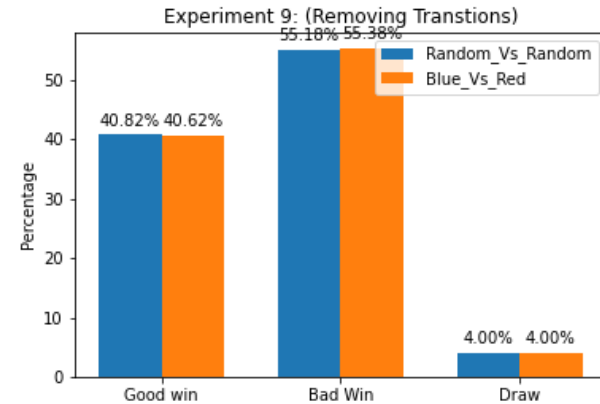
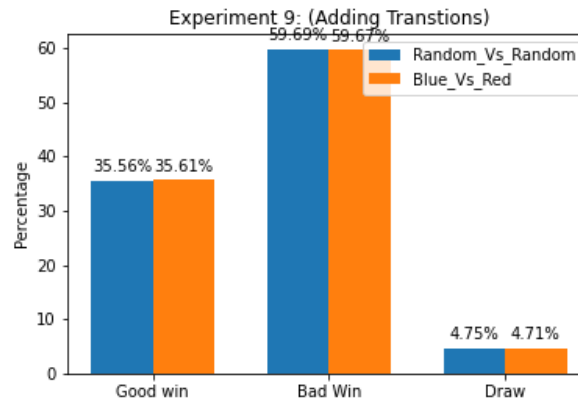
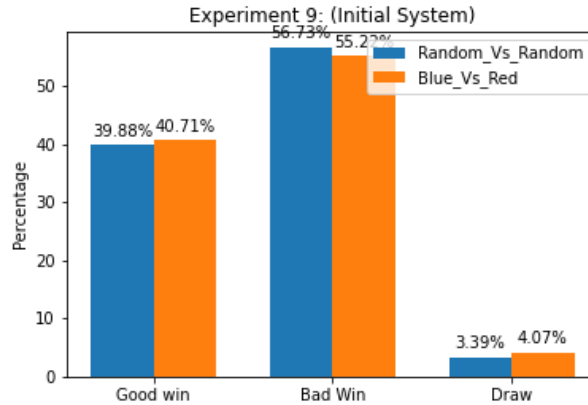
- Future Work:

- Different **RL algorithms** other than can be used and compared to train the agents such as DQN, DDPG.
- Different **optimization techniques** can be used to optimize the agents' initial resources.

Conclusion and Future Work ○ ○

Experiment: Transitions Add/Remove

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References

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- ✓ Path Elimination Algorithm
- ✓ Conclusion and Future Work
- **References**

- [1] *Model Based Systems Engineering (MBSE) with Enterprise Architect* | Sparx Systems. Model Based Systems Engineering (MBSE) With Enterprise Architect | Sparx Systems. Retrieved May 30, 2023, from <https://sparxsystems.com/mbse/>
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Thank you!



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